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3 rows 1919s runtime Refreshed at 9/26/26, 9:05 PM (25m ago)

```
def to_bronze(name):
    df = (spark.read.option("header", True).csv(f"/V01/{name}/raw/{name}")
        .withColumn("source_file", F.col("metadata.file_path"))
        .withColumn("_ingested_at", F.current_timestamp()))
    df.write.mode("overwrite").saveAsTable(f"bronze_{name}_{HW_ID}")
    return df

bronze_stores = to_bronze("stores")
bronze_footfall = to_bronze("footfall")
bronze_bills = to_bronze("bills")

> See performance (3) ✓ Checked for improvements
3 dataframes: bronze_bills, bronze_footfall, bronze_stores
```

```
print("footfall rows:", spark.table("bronze_footfall_{HW_ID}").count())
print("stores rows:", spark.table("bronze_stores_{HW_ID}").count())
print("bills rows:", spark.table("bronze_bills_{HW_ID}").count())

> See performance (3) ✓ Checked for improvements
footfall rows: 12960
stores rows: 12
bills rows: 86659
```

```
from pyspark.sql.window import Window

silver_footfall = (bronze_footfall
    .withColumn("footfall", F.expr("try_cast(footfall as int)"))
    .withColumn("footfall", F.when(F.col("counter_status")=="OFFLINE", None).otherwise(F.col("footfall")))
    .withColumn("trade_date", F.expr("try_cast(trade_date as date)"))
    .withColumn("sensor_ok", F.col("footfall").isNotNull()))

w = Window.partitionBy("bill_id").orderBy(F.col("_ingested_at").desc())
silver_bills = (bronze_bills
    .withColumn("bill_amount", F.expr("try_cast(bill_amount as double)"))
    .withColumn("rev", F.row_number().over(w).filter("rev > 1").drop(True))
    .join(bronze_stores.select("store_id", "store_id", "inner")
    .withColumn("hr", F.hour("bill_ts"))
    .filter((F.col("hr") >= 10) & (F.col("hr") <= 21)))

silver_footfall.write.mode("overwrite").saveAsTable("silver_footfall_{HW_ID}")
silver_bills.write.mode("overwrite").saveAsTable("silver_bills_{HW_ID}")

> See performance (2) ✓ Checked for improvements
2 dataframes: silver_bills, silver_footfall
```

```
print("bronze bills:", bronze_bills.count())
print("silver bills:", spark.table("silver_bills_{HW_ID}").count())
print("difference:", bronze_bills.count() - spark.table("silver_bills_{HW_ID}").count())

> See performance (4) ✓ Checked for improvements
bronze bills: 86659
silver bills: 86594
difference: 265
```

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```
from pyspark.sql.window import Window

silver_footfall = (bronze_footfall
    .withColumn("footfall", F.expr("try_cast(footfall as int)"))
    .withColumn("footfall", F.when(F.col("counter_status")=="OFFLINE", None).otherwise(F.col("footfall")))
    .withColumn("trade_date", F.expr("try_cast(trade_date as date)"))
    .withColumn("sensor_ok", F.col("footfall").isNotNull()))

w = Window.partitionBy("bill_id").orderBy(F.col("_ingested_at").desc())
silver_bills = (bronze_bills
    .withColumn("bill_amount", F.expr("try_cast(bill_amount as double)"))
    .withColumn("rev", F.row_number().over(w).filter("rev > 1").drop(True))
    .join(bronze_stores.select("store_id", "store_id", "inner")
    .withColumn("hr", F.hour("bill_ts"))
    .filter((F.col("hr") >= 10) & (F.col("hr") <= 21)))

silver_footfall.write.mode("overwrite").saveAsTable("silver_footfall_{HW_ID}")
silver_bills.write.mode("overwrite").saveAsTable("silver_bills_{HW_ID}")

> See performance (2) ✓ Checked for improvements
2 dataframes: silver_bills, silver_footfall
```

```
print("bronze bills:", bronze_bills.count())
print("silver bills:", spark.table("silver_bills_{HW_ID}").count())
print("difference:", bronze_bills.count() - spark.table("silver_bills_{HW_ID}").count())

> See performance (4) ✓ Checked for improvements
bronze bills: 86659
silver bills: 86594
difference: 265
```

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```
gold = (silver_footfall.join(
    silver_bills.groupBy("store_id", F.to_date("trade_date"), F.hour("bill_ts").alias("hour"))
    .agg(F.count("*").alias("bills"), F.sum("bill_amount").alias("revenue")),
    ["store_id", "trade_date", "hour", "left"])
    .join(bronze_stores.select("store_id", "city", "format", "store_id", "left")
    .withColumn("bills", F.coalesce("bills", F.lit(0)))
    .withColumn("revenue", F.coalesce("revenue", F.lit(0.0)))
    .withColumn("conversion_rate", F.when(F.col("sensor_ok"), F.col("bills")/F.col("footfall"))))
    .write.mode("overwrite").saveAsTable("gold_store_hour_{HW_ID}")

> See performance (1) ✓ Checked for improvements
gold: pyspark.sql.connect.dataframe.DataFrame = [store_id:string, trade_date:date ... 11 more fields]
```

```
print("gold rows:", gold.count())
print("sensor offline rows:", gold.filter("sensor_ok = false").count())

> See performance (2) ✓ Checked for improvements
gold rows: 12960
sensor offline rows: 540
```

```
(gold.coalesce(1)).write.mode("overwrite").option("header", True)
.csv(f"/V01/export/gold_store_hour")

> See performance (1) ✓ Checked for improvements
```

app.snowflake.com/ap-southeast-7.aws/mw47097/#/workspaces/ws/USER%24/PUBLIC/DEFAULT%24/Untitled.sql

Workspaces Databases +

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Untitled.sql

```
5 USE DATABASE CAPSTONE_DB;
6 CREATE STAGE IF NOT EXISTS CAPSTONE_STAGE;
7 CREATE OR REPLACE TABLE GOLD_STORE_HOUR (
8   store_id STRING,
9   trade_date DATE,
10  hour INT,
11  counter_status STRING,
12  footfall INT,
13  source_file STRING,
14  _ingested_at TIMESTAMP_NTZ,
15  sensor_ok BOOLEAN,
16  bills INT,
17  revenue DOUBLE,
18  city STRING,
19  format STRING,
20  conversion_rate DOUBLE
21 );
22
23 COPY INTO GOLD_STORE_HOUR
24 FROM CAPSTONE_STAGE
25 FILE_FORMAT = (TYPE=CSV SKIP_HEADER=1 NULL_IF=('', NULL));
26
27 COPY INTO GOLD_STORE_HOUR
28 FROM CAPSTONE_STAGE
29 FILE_FORMAT = (TYPE=CSV SKIP_HEADER=1 NULL_IF=('', NULL));
30 SELECT hour, ROUND(SUM(bills)/NULLIF(SUM(footfall),0), 4) AS conversion_rate
31 FROM GOLD_STORE_HOUR WHERE sensor_ok
32 GROUP BY hour ORDER BY conversion_rate ASC LIMIT 3;
```

Results (just now)

db	file	status	rows_parsed	rows_loaded	error_limit	errors_seen	first_error	first_error_line	first_error_character	first_error_column_name
1	capstone_stage/gold_st	LOADED	12960	12960	1	0	null	null	null	null

app.snowflake.com/ap-southeast-7.aws/mw47097/#/workspaces/ws/USER%24/PUBLIC/DEFAULT%24/Untitled.sql

Workspaces Databases +

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Untitled.sql10 hour INT,
11 counter_status STRING,
12 footfall INT,
13 source_file STRING,
14 _ingested_at TIMESTAMP_NTZ,
15 sensor_ok BOOLEAN,
16 bills INT,
17 revenue DOUBLE,
18 city STRING,
19 format STRING,
20 conversion_rate DOUBLE
21);
22
23 COPY INTO GOLD_STORE_HOUR
24 FROM CAPSTONE_STAGE
25 FILE_FORMAT = (TYPE=CSV SKIP_HEADER=1 NULL_IF=('', NULL));
26
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30 SELECT hour, ROUND(SUM(bills)/NULLIF(SUM(footfall),0), 4) AS conversion_rate
31 FROM GOLD_STORE_HOUR WHERE sensor_ok
32 GROUP BY hour ORDER BY conversion_rate ASC LIMIT 3;

Results (just now)

db	status
1	Copy executed with 0 files processed.

app.snowflake.com/ap-southeast-7.aws/mw47097/#/workspaces/ws/USER%24/PUBLIC/DEFAULT%24/Untitled.sql

Workspaces Databases +

My Workspace + Add new

Untitled.sql1 CREATE DATABASE IF NOT EXISTS CAPSTONE_DB WITH WAREHOUSE=SEED;
2 CREATE DATABASE IF NOT EXISTS CAPSTONE_DB;
3 CREATE SCHEMA IF NOT EXISTS CAPSTONE_DB.PUBLIC;
4 USE DATABASE CAPSTONE_DB;
5 USE SCHEMA CAPSTONE_DB;
6 CREATE STAGE IF NOT EXISTS CAPSTONE_STAGE;
7 CREATE OR REPLACE TABLE GOLD_STORE_HOUR (
8 store_id STRING,
9 trade_date DATE,
10 hour INT,
11 counter_status STRING,
12 footfall INT,
13 source_file STRING,
14 _ingested_at TIMESTAMP_NTZ,
15 sensor_ok BOOLEAN,
16 bills INT,
17 revenue DOUBLE,
18 city STRING,
19 format STRING,
20 conversion_rate DOUBLE
21);
22
23 COPY INTO GOLD_STORE_HOUR
24 FROM CAPSTONE_STAGE
25 FILE_FORMAT = (TYPE=CSV SKIP_HEADER=1 NULL_IF=('', NULL));
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27 COPY INTO GOLD_STORE_HOUR
28 FROM CAPSTONE_STAGE
29 FILE_FORMAT = (TYPE=CSV SKIP_HEADER=1 NULL_IF=('', NULL));
30 SELECT hour, ROUND(SUM(bills)/NULLIF(SUM(footfall),0), 4) AS conversion_rate
31 FROM GOLD_STORE_HOUR WHERE sensor_ok
32 GROUP BY hour ORDER BY conversion_rate ASC LIMIT 3;

Results (1 minute ago)

db	hour	conversion_rate
1	15	0.0863
2	16	0.0847
3	14	0.0801